

Customer Behavior Analysis

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Project: Customer Behavior Analysis

Organization: Alfido Tech Data Analyst Internship

Executive Summary

The **Customer Behavior Analysis** project was carried out as part of the **Alfido Tech Data Analyst Internship** to analyze customer transaction data and gain meaningful business insights. The primary objective of this project was to understand customer purchasing behavior, identify customer segments, analyze purchase patterns, and recognize customers who may be at risk of churning. By applying data analysis techniques and visualization, the project demonstrates how data can support informed business decisions and improve overall customer relationship management.

The analysis was performed using **Python** along with libraries such as **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn** within a Jupyter Notebook environment. The dataset was first cleaned by checking for missing values, removing duplicate records, and converting data into appropriate formats. This preprocessing ensured that the analysis was accurate and reliable.

Exploratory Data Analysis (EDA) was conducted to understand customer demographics, purchasing behavior, payment preferences, revenue distribution, product category performance, returns, and customer churn. Several visualizations were created to present these insights clearly, including charts for gender distribution, age distribution, product categories, payment methods, monthly revenue trends, revenue by category, customer churn, and correlation analysis. These visualizations made it easier to identify important patterns and trends within the dataset.

To better understand customer value, **RFM (Recency, Frequency, Monetary) Analysis** was performed. Customers were segmented into four categories: **High Value**, **Loyal**, **Regular**, and **At Risk**. High Value customers were identified as the most profitable and engaged customers, while At Risk customers represented individuals who had reduced their purchasing activity and may require retention efforts. This segmentation enables businesses to design targeted marketing campaigns and improve customer engagement strategies.

The project also identified several important business findings. High-value customers contribute significantly to total revenue, making customer retention strategies essential. Revenue patterns varied across months, indicating seasonal purchasing behavior that businesses can use for planning promotions and inventory. Product category analysis highlighted the best-performing categories, while payment method analysis revealed customer preferences that can improve the purchasing experience. Additionally, churn analysis emphasized the importance of early customer engagement to reduce customer loss.

Based on these findings, several business recommendations were proposed. These include implementing loyalty reward programs for high-value customers, developing personalized

marketing campaigns for at-risk customers, promoting top-performing product categories, improving product quality to reduce returns, and using customer segmentation to deliver personalized recommendations and offers. These strategies can help increase customer satisfaction, strengthen customer loyalty, and improve overall business performance.

Overall, this project successfully demonstrates the practical application of data analysis, visualization, and customer segmentation techniques to solve real-world business problems. The insights generated from this analysis provide valuable support for strategic decision-making and highlight the importance of data-driven approaches in enhancing customer relationships and maximizing business growth.

Customer Segments

- **High Value:** Customers with recent, frequent, and high-value purchases.
 - **Loyal:** Customers who purchase consistently over time.
 - **Regular:** Customers with moderate purchasing activity and future growth potential.
 - **At Risk:** Customers with low recent activity who may stop purchasing if not re-engaged.
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Key Findings

- Customer purchasing behavior differs across product categories.
 - Monthly revenue analysis indicates seasonal purchasing trends.
 - High-value customers contribute significantly to business revenue.
 - Customer segmentation effectively identifies loyal and at-risk customers.
 - Payment method analysis provides insights into customer preferences.
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Business Recommendations

- Introduce loyalty reward programs for high-value customers.
 - Launch personalized marketing campaigns for at-risk customers.
 - Promote top-performing product categories to increase sales.
 - Improve product quality and customer support to reduce returns.
 - Use customer segmentation to deliver targeted promotions and improve customer retention.
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Project Links

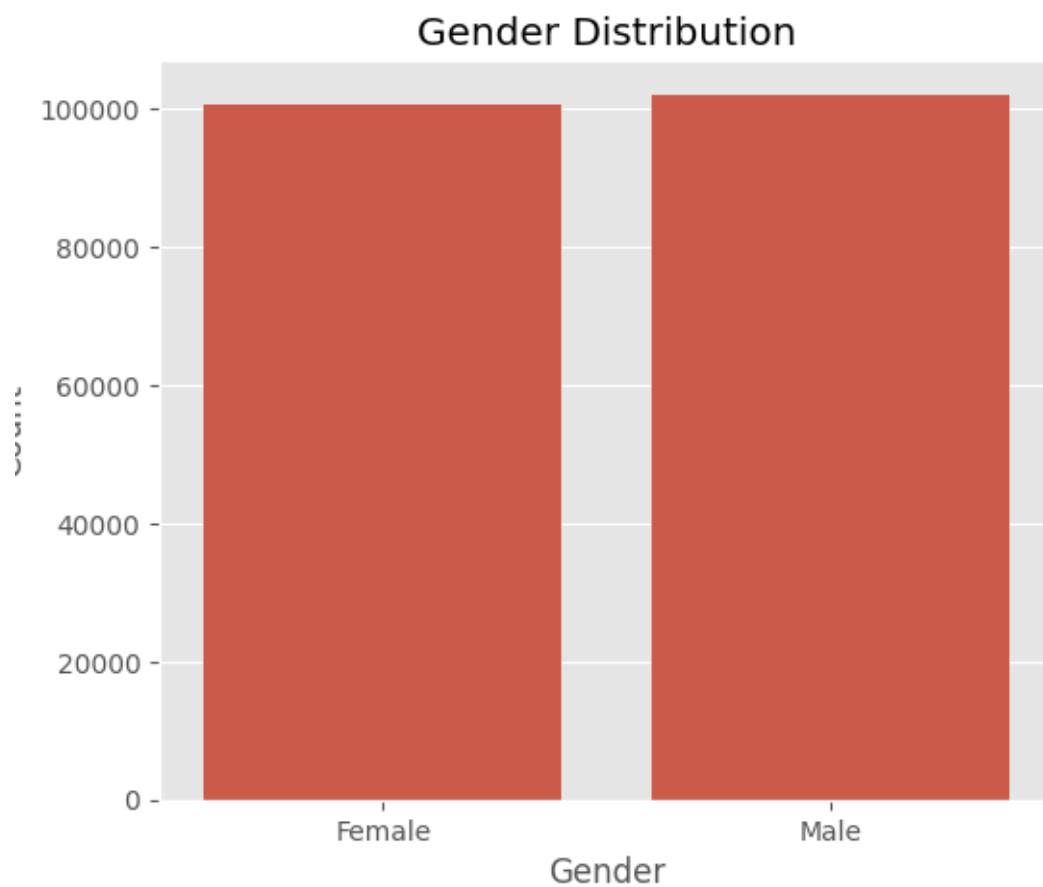
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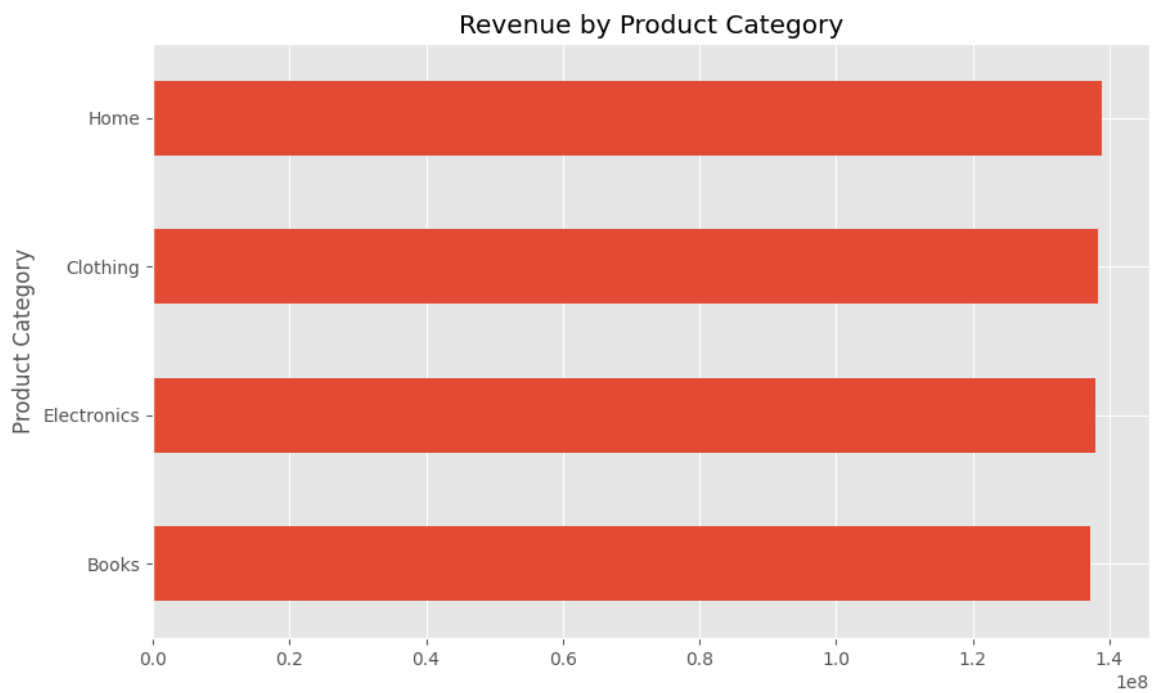
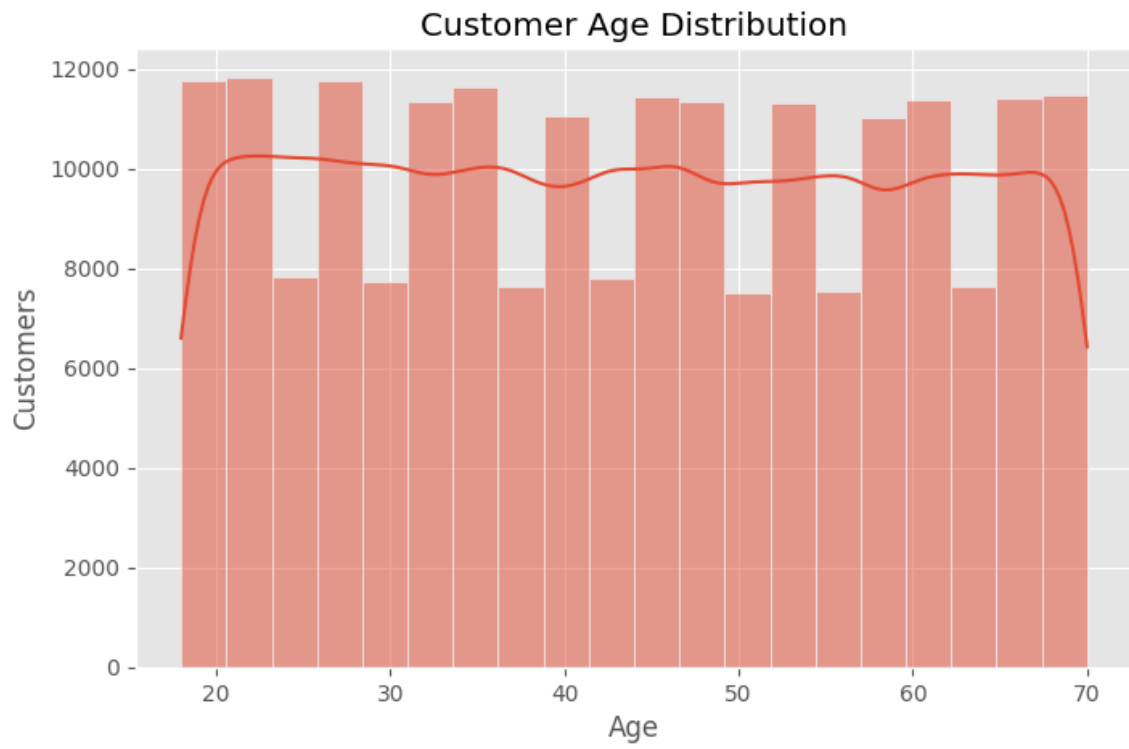
<https://github.com/SuryaTeja525/Customer-Behavior-Analysis>

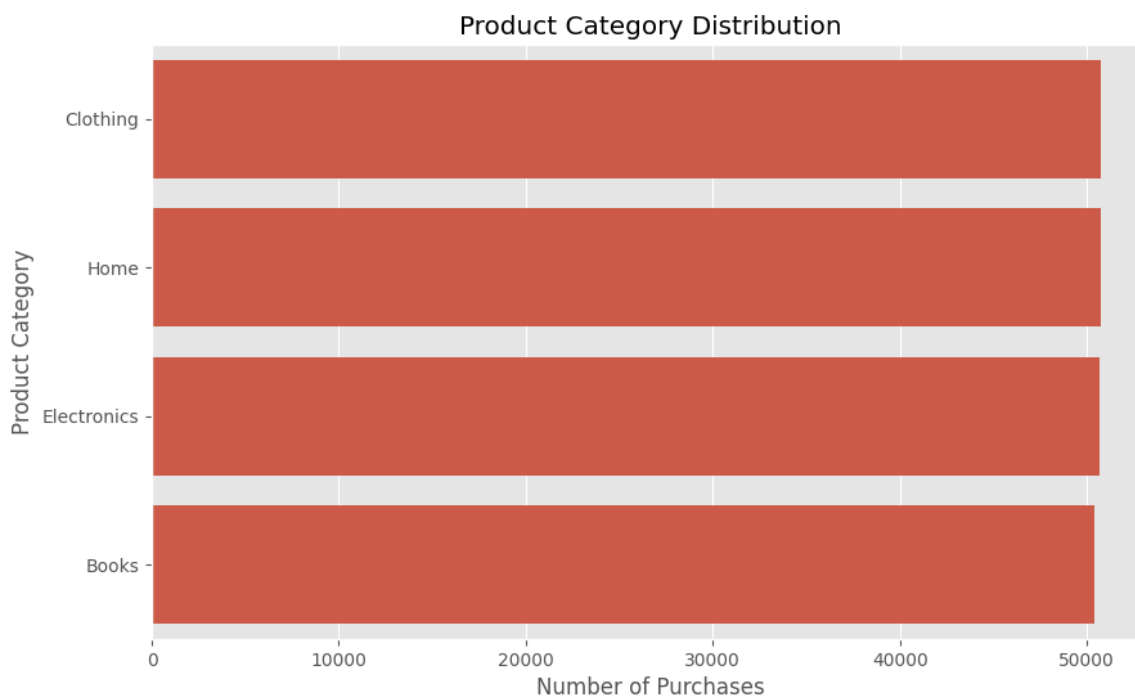
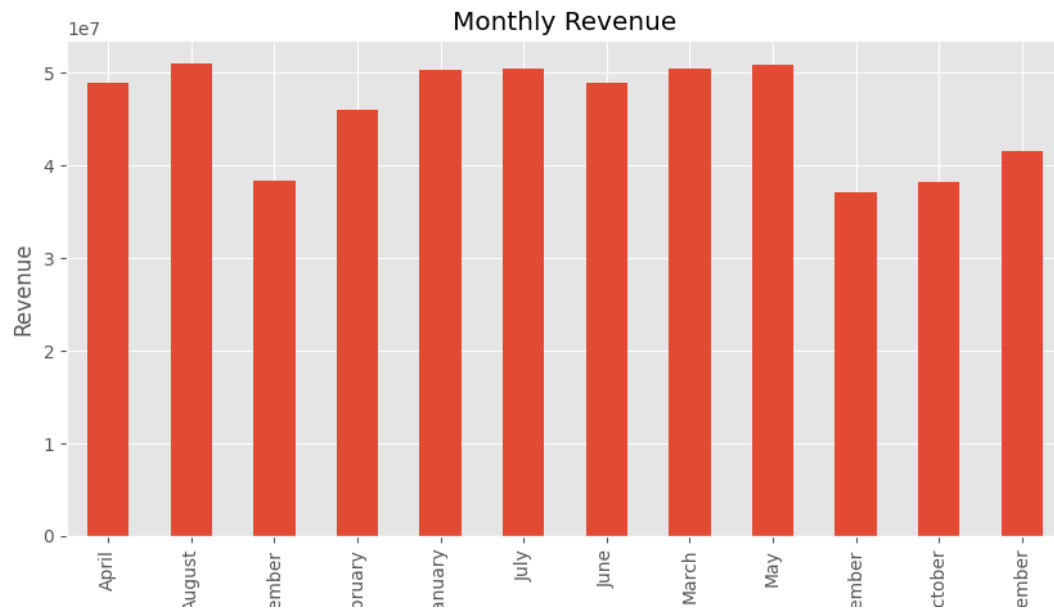
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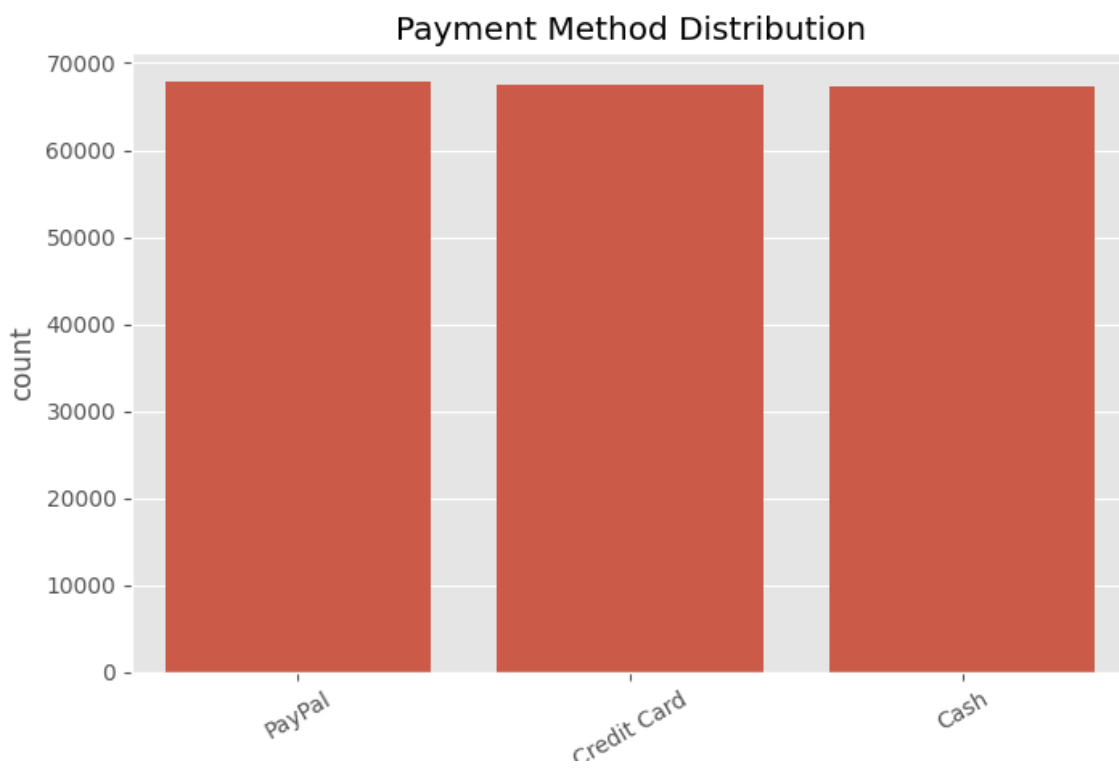
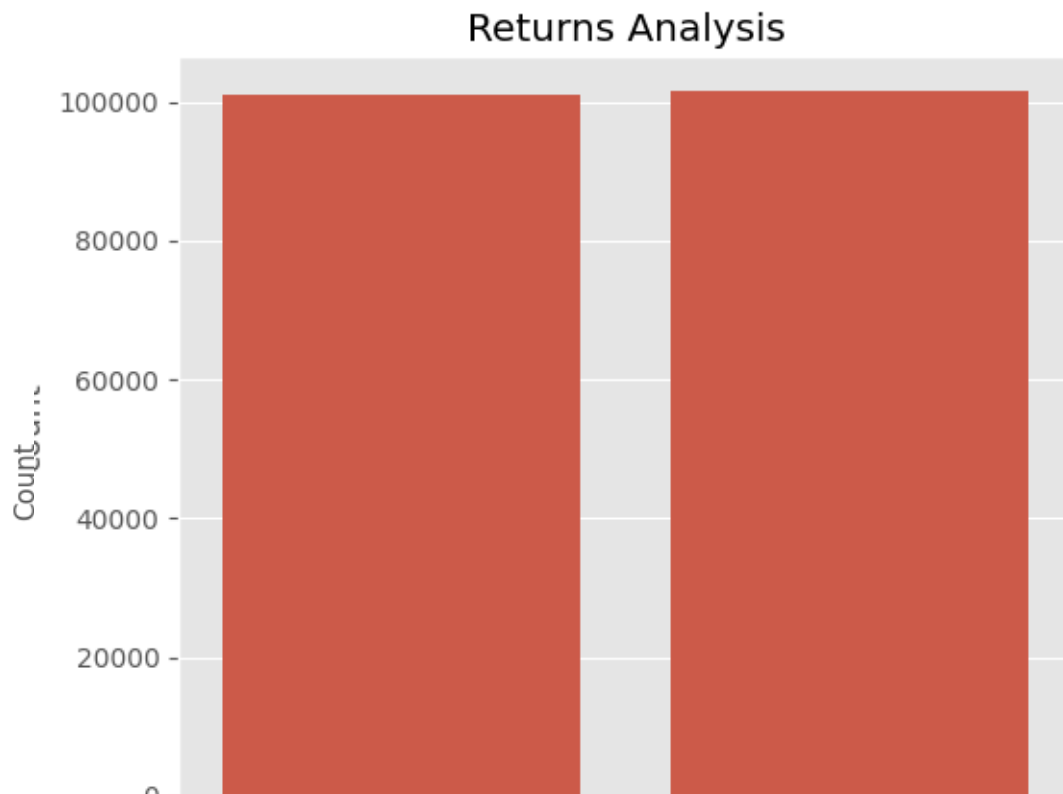
<https://www.kaggle.com/datasets/bhanupratapbiswas/customer-behavior-analysis>

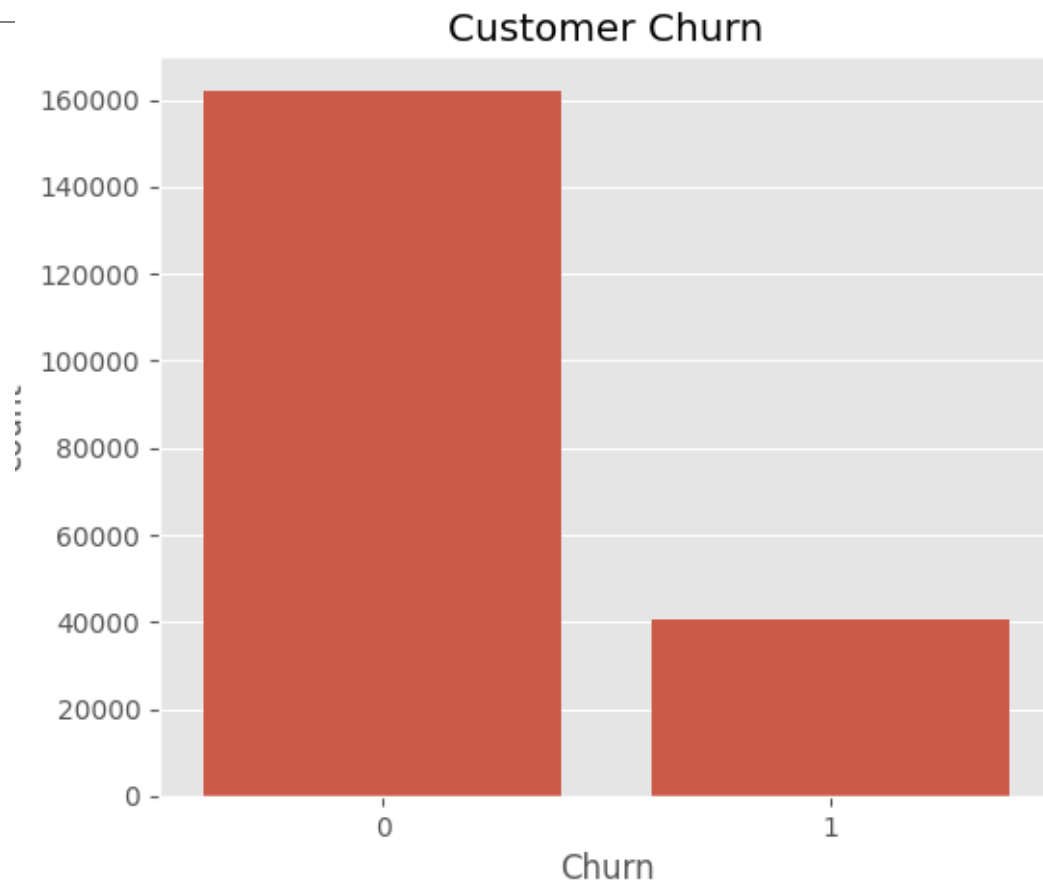
Visualizations











Conclusion

This project successfully analyzed customer transaction data, identified purchasing patterns, segmented customers using RFM Analysis, and generated actionable business recommendations. The findings demonstrate how data analysis techniques can support strategic business decisions, improve customer retention, and enhance overall business performance.